

## ► MICROBIOLOGY—MYCOLOGY

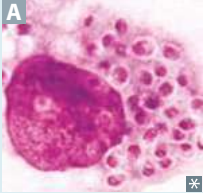
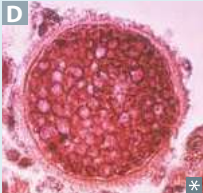
**Systemic mycoses**

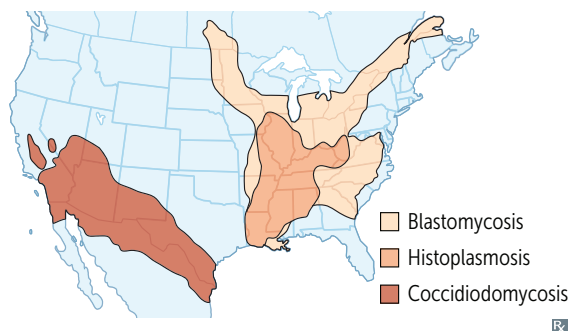
All of the following can cause pneumonia and can disseminate.

All are caused by dimorphic fungi: **cold** (20°C) = **mold**; **heat** (37°C) = **yeast**. Only exception is *Coccidioides*, which is a spherule (not yeast) in tissue.

Systemic mycoses can form granulomas (like TB); cannot be transmitted person-to-person (unlike TB).

Treatment: fluconazole or itraconazole for **local** infection; amphotericin B for **systemic** infection.

| DISEASE                                                                                                               | ENDEMIC LOCATION                    | PATHOLOGIC FEATURES                                                                                                | UNIQUE SIGNS/SYMPTOMS                                                                                                                 | NOTES                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Histoplasmosis</b><br>            | Mississippi and Ohio River Valleys  | Macrophage filled with <i>Histoplasma</i> (smaller than RBC) <b>A</b><br>Tuberculate macroconidia on culture       | Palatal/tongue ulcers, splenomegaly, pancytopenia, erythema nodosum                                                                   | <b>Histo hides</b> (within macrophages)<br>Associated with bird or bat droppings (eg, caves)<br>Diagnosis via urine/serum antigen |
| <b>Blastomycosis</b><br>            | Eastern and Central US, Great Lakes | <b>Broad</b> -based budding of <i>Blastomyces</i> (same size as RBC) <b>B</b>                                      | Inflammatory lung disease<br>Disseminates to bone/skin (verrucous lesions <b>C</b> , may mimic SCC).                                  | <b>Blasto buds broadly</b><br>                |
| <b>Coccidioidomycosis</b><br>      | Southwestern US, California         | Spherule filled with endospores of <i>Coccidioides</i> (much larger than RBC) <b>D</b>                             | Disseminates to bone/skin<br>Erythema nodosum (desert bumps) or multiforme<br>Arthralgias (desert rheumatism)<br>Can cause meningitis | Associated with dust exposure in endemic areas (eg, archeological excavations, earthquakes)                                       |
| <b>Para-coccidioidomycosis</b><br> | <b>Latin America</b>                | Budding yeast of <i>Paracoccidioides</i> with “ <b>captain’s wheel</b> ” formation (much larger than RBC) <b>E</b> | Similar to blastomycosis, males > females                                                                                             | <b>Paracoccidio parasails</b> with the <b>captain’s wheel</b> all the way to <b>Latin America</b>                                 |



### Opportunistic fungal infections

#### *Candida albicans*

*alba* = white. Dimorphic; forms pseudohyphae and budding yeasts at 20°C **A**, germ tubes at 37°C **B**.

Systemic or superficial fungal infection. Causes oral **C** and esophageal thrush in immunocompromised (neonates, steroids, diabetes, AIDS), vulvovaginitis (diabetes, use of antibiotics), diaper rash, infective endocarditis (people who inject drugs), disseminated candidiasis (especially in neutropenic patients as host defense relies on phagocytes), chronic mucocutaneous candidiasis.

Treatment: oral fluconazole/topical azoles for vaginal; nystatin, azoles, or, rarely, echinocandins for oral; fluconazole, echinocandins, or amphotericin B for esophageal or systemic disease.

#### *Aspergillus fumigatus*

**A**cute angle (45°) **D** branching of **s**eptate hyphae.

Causes invasive aspergillosis in immunocompromised patients, especially those with neutrophil dysfunction (eg, chronic granulomatous disease) because *Aspergillus* is catalase ⊕.

Can cause aspergillomas **E** in pre-existing lung cavities, especially after TB infection.

Some species of *Aspergillus* produce aflatoxins (induce TP53 mutations leading to hepatocellular carcinoma).

Treatment: voriconazole or echinocandins (2nd-line).

**Allergic bronchopulmonary aspergillosis (ABPA)**—hypersensitivity response to *Aspergillus* growing in lung mucus. Associated with asthma and cystic fibrosis; may cause bronchiectasis and eosinophilia.

#### *Cryptococcus neoformans*

5–10 μm with narrow budding. Heavily encapsulated yeast. Not dimorphic. ⊕ PAS staining.

Found in soil, pigeon droppings. Acquired through inhalation with hematogenous dissemination to meninges. Highlighted with India ink (clear halo **F**) and mucicarmine (red inner capsule **G**).

Latex agglutination test detects polysaccharide capsular antigen and is more sensitive and specific. Causes cryptococcosis, which can manifest with meningitis, pneumonia, and/or encephalitis (“soap bubble” lesions in brain), primarily in immunocompromised.

Treatment: amphotericin B + flucytosine followed by fluconazole for cryptococcal meningitis.

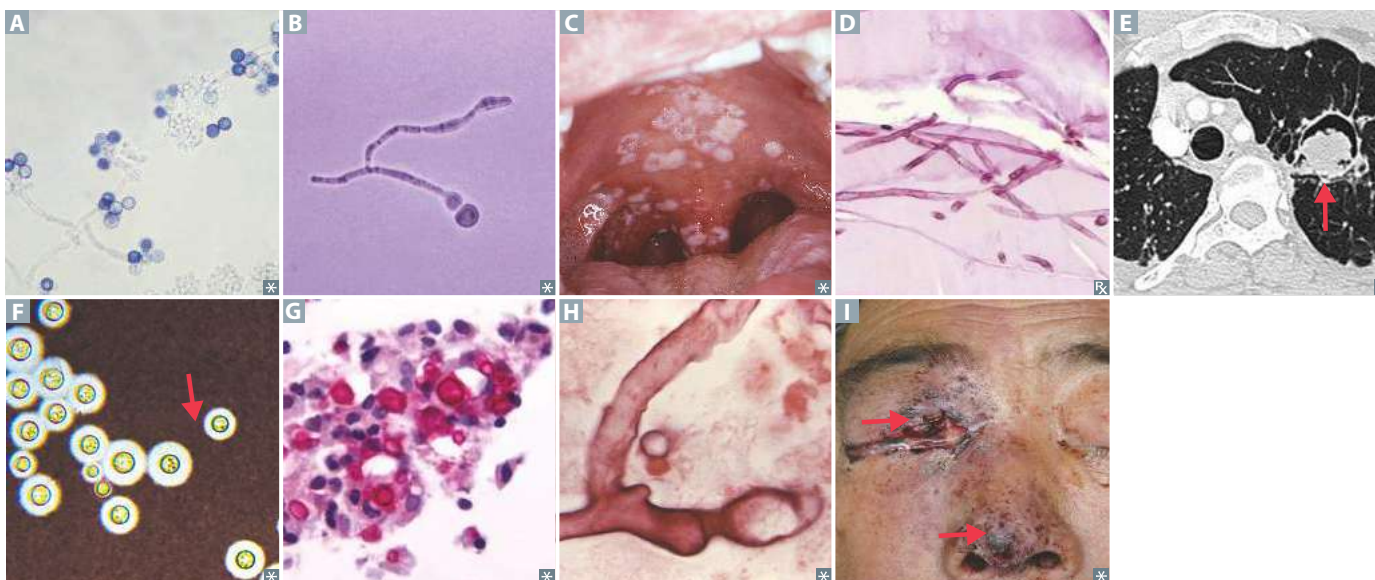
#### *Mucor and Rhizopus* spp

Irregular, broad, nonseptate hyphae branching at wide angles **H**.

Causes mucormycosis, mostly in patients with DKA and/or neutropenia (eg, leukemia). Inhalation of spores → fungi proliferate in blood vessel walls, penetrate cribriform plate, and enter brain.

Rhinocerebral, frontal lobe abscess; cavernous sinus thrombosis. Headache, facial pain, black necrotic eschar on face **I**; may have cranial nerve involvement.

Treatment: surgical debridement, amphotericin B or isavuconazole.



***Pneumocystis jirovecii***

Yeastlike fungus that causes *Pneumocystis* pneumonia (PCP), a diffuse interstitial pneumonia **A**. Most infections are asymptomatic. Immunosuppression (eg, AIDS) predisposes to disease. Diffuse, bilateral ground-glass opacities on chest imaging, with pneumatoceles **B**. Diagnosed by bronchoalveolar lavage or lung biopsy. Disc-shaped yeast seen on methenamine silver stain of lung tissue **C** or with fluorescent antibody.

Treatment/prophylaxis: TMP-SMX, pentamidine, dapsone (prophylaxis as single agent, or treatment in combination with TMP), atovaquone. Start prophylaxis when CD4+ cell count drops to  $< 200$  cells/mm<sup>3</sup> in people living with HIV.

***Sporothrix schenckii***

Causes sporotrichosis. Dimorphic fungus. Exists as a **cigar**-shaped yeast at 37 °C in the human body and as hyphae with spores in soil (conidia). Lives on vegetation. When spores are traumatically introduced into the skin, typically by a thorn ("**rose gardener's** disease"), causes local pustule or ulcer with nodules along draining lymphatics (ascending lymphangitis **A**).

Disseminated disease possible in immunocompromised host.

Treatment: itraconazole or **pot**assium iodide (only for cutaneous/lymphocutaneous).

Think of a **rose gardener** who smokes a **cigar** and **pot**.